

3. MICROTOME

Production of thin sections from paraffin-embedded tissue blocks for microscopic examination.

SOP No.: LAB-HISTO-03	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To cut reproducible thin sections, commonly in the laboratory's validated thickness range, from paraffin blocks and mount them on microscope slides.

PPE & Safety

- Use cut-resistant handling practices and never touch the blade edge. Keep fingers away from the blade and specimen clamp. Lock the handwheel before blade adjustment, cleaning or changing blocks/blades. Dispose of blades only in an approved sharps container.

Materials

- Paraffin block; microtome; disposable blade/approved knife; forceps; water bath if used; charged microscope slides; brush; blade guard.

Procedure

- Inspect the microtome, ensure the handwheel is locked during setup, and install the blade according to manufacturer instructions.
- Clamp the block securely and align it. Set the required section thickness using the validated laboratory setting.
- Remove excess wax by coarse trimming with the handwheel locked/unlocked only as specified by the manufacturer.
- Unlock the handwheel and section smoothly. Collect ribbons using forceps/brush without contacting the blade.
- Float sections on the laboratory-approved water bath if used; pick up suitable sections onto labelled slides.
- Drain/dry slides using the validated method before staining. Re-lock the handwheel whenever the blade area is accessed.

Quality Control

- Sections should be continuous, even and free from chatter, compression, folds and knife lines. Confirm tissue is present and correctly identified.

Cleaning & Shutdown

- Lock the handwheel, lower/guard the blade and remove/dispose of the blade safely. Brush away wax debris without contacting the blade. Clean the work area and return the instrument to its designated safe state.

Troubleshooting

- Compression → check block hardness, blade condition and clearance. Chatter → check clamping, blade stability and block support. Folds → adjust water-bath conditions/section handling. Knife lines → replace damaged/contaminated blade.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.